



ORIGINAL ARTICLE

Measurements of gastric emptying during continuous nasogastric infusion of liquid feed: Electric impedance tomography versus gamma scintigraphy[☆]

Clare T. Soulsby^a, Mandeep Khela^a, Etsuro Yazaki^a, David F. Evans^a,
Enid Hennessy^b, Jeremy Powell-Tuck^{a,*}

^aCentre for Adult and Paediatric Gastroenterology, Institute for Cell and Molecular Science,
Barts and The London Medical School, London, UK

^bStatistics Wolfson Institute of Preventive Medicine, Barts and The London Medical School, London, UK

Received 29 June 2005; accepted 30 November 2005

KEYWORDS

Enteral feed;
Continuous
nasogastric infusion;
Electric impedance
tomography (EIT);
Gamma scintigraphy;
Gastric residual
volume (GRV)

Summary

Background & aims: Continuous nasogastric infusion is commonly used to deliver enteral feed but current methods used to assess tolerance based on aspiration and measurement of gastric residual volume have been criticised. Electric impedance tomography (EIT) measures gastric emptying by monitoring changes in epigastric impedance when a meal progressively empties from the stomach. Aims: (1) to establish whether EIT was a valid method for measuring gastric emptying during continuous nasogastric infusion by comparing it with gamma scintigraphy (GS) and (2) to provide data on gastric emptying patterns during continuous nasogastric infusion.

Methods: Gastric emptying of 400 ml of enteral feed given over 200 min was measured simultaneously using EIT and GS in 10 healthy volunteers (five male and five female).

Results: Gastric emptying curves were obtained in 10 subjects by EIT but only eight by GS. Visual examination of the curves showed reasonable agreement. Patterns of emptying and filling during continuous nasogastric infusion were variable between individuals; the prevailing pattern was a trend towards a steady-state volume of approximately 50–125 ml.

[☆]Supported by grants from NHS Research and Development and Nutricia Clinical Care.

*Corresponding author. Wingate Institute, Royal London Hospital, 26 Ashfield Street, London E1 2AJ, UK. Tel.: +44 20 7882 2631; fax: +44 20 7375 2031.

E-mail addresses: c.t.soulsby@qmul.ac.uk (C.T. Soulsby), j.powell-tuck@qmul.ac.uk (J. Powell-Tuck).

Conclusions: While EIT does not provide an accurate estimate of gastric volume during continuous infusion, it does show patterns of gastric emptying over time. With further development this could make it a useful tool for monitoring gastric emptying in patients at risk of gastroparesis.

© 2006 Elsevier Ltd and European Society for Clinical Nutrition and Metabolism. All rights reserved.

Introduction

Continuous nasogastric infusion is the most commonly used method for delivering enteral feed to patients in UK hospitals, but there is little information on the patterns of gastric emptying during such infusion. The method currently used to assess patient tolerance of continuous nasogastric infusion of enteral feed is 4–6 hourly aspiration of gastric contents and measurement of the gastric residual volume (GRV); a volume greater than 150–200 ml is considered indicative of poor enteral feed tolerance due to delayed gastric emptying.

Assessment of GRV by aspiration has been criticised because it does not correlate with physical examination and radiographic results¹ and the volume aspirated depends on the position of both the tube and the patient.² Furthermore, there appears to be little evidence to support the 150–200 ml cut off currently advocated as it does not take into consideration saliva or gastric secretions, which contribute up to 4500 ml/day.³ Due to the paucity of data on gastric emptying patterns during feed infusion authors have used mathematical models, which predict steady-state gastric volumes of between 137 and 649 ml.^{3,4}

The development of a reliable and valid method that could provide real-time images of gastric emptying patterns and gastric volume during continuous nasogastric infusion would be beneficial as it would:

1. allow enteral feed tolerance to be assessed during feed infusion on an individual patient basis,
2. provide data on gastric emptying patterns during continuous infusion and allow the development of more evidence based feed delivery protocols and cut-off volumes.

Electric impedance tomography (EIT) is a non-invasive technique which has primarily been developed to measure gastric emptying of food given as a meal or "bolus". Transfer impedance is measured by applying current to electrodes placed around

the abdomen. Gastric emptying is measured by recording impedance changes in the stomach when a meal of fixed resistivity progressively empties. Results are usually expressed as gastric emptying half time ($t_{1/2}$). EIT has been shown to provide accurate measurements of volume changes in vitro, and in vivo EIT accurately measured volume changes in balloons placed in the stomach.⁵ EIT correlates with gastric emptying measured by dye dilution and has been validated by comparison with gamma scintigraphy (GS), as a method for measuring gastric emptying of liquid boluses⁵ and solid meals^{5–7} (Table 1). Not only is the system non-invasive, but it is also small and portable so offers the potential for bedside measurement of gastric emptying. Because of the non-invasive, portable nature of EIT, we wished to use it to investigate gastric emptying and enteral feed tolerance in critically ill patients. However, as many patients are fed using continuous nasogastric infusion, it was necessary to establish whether EIT was a valid method for measuring gastric emptying of enteral feed when it was administered continuously.

To validate EIT, we chose to compare it with GS as the "gold standard".^{8,9} GS measures gastric emptying by tracking the passage of a radio nuclide labelled meal as it moves out of the stomach, using a gamma camera. Despite widely being accepted as the gold standard, GS does have a number of disadvantages (Table 2) which should be taken into consideration when comparing it with EIT. As EIT measures total gastric volume, that is the test meal and gastric secretions, and GS only measures the radio marker, these methods only agree when gastric secretions are inhibited.

Aims

- (1) To establish whether EIT is a valid method for measuring gastric emptying during continuous nasogastric infusion of enteral feed.
- (2) To provide the first published data on gastric emptying patterns during continuous nasogastric infusion.

Table 1 Studies investigating validity of EIT to measure gastric emptying of EIT given as a bolus.

Study	Aim of study	Subjects	Test meal	Acid suppression	Methodology	Results
5	Effect of acid inhibition on repeatability of GE measured by EIT	8 normals	1 oxo cube plus water	No inhibition vs. 800 mg cimetidine	GE measured by EIT on four occasions per subject, two consecutive days with acid inhibition or placebo in randomised order	Good repeatability for $t_{1/2}^1$ with acid suppression ($r = 0.90$) poor repeatability without ($r = -0.19$)
6	Effect of acid inhibition on accuracy of GE measured by EIT compared with scintigraphy	20 normals	160 g beefburger	No inhibition vs. 800 mg cimetidine	GE measured simultaneously by EIT vs. scintigraphy with acid inhibition ($n = 12$) or without ($n = 8$)	Good agreement for $t_{1/2}^1$ ($r = 0.713$), lag time ($r = 0.585$) with acid inhibition. Poor agreement $t_{1/2}^1$ ($r = 0.058$), lag time ($r = 0.376$) without acid inhibition
7	Effect of different types of acid inhibitors on repeatability of GE measured by EIT	16 normals	1 oxo cube plus water	No inhibition vs. 800 mg cimetidine vs. 40 mg omeprazole	GE measured by EIT on three occasions per subject, once with no acid inhibition, once with cimetidine, once with omeprazole	No differences between males and females Acid inhibition increased speed of $t_{1/2}^1$ emptying ($P = 0.06$ cimetidine, $P = 0.09$ omeprazole)
7	Effect of different types of acid inhibitors on repeatability of GE measured by EIT	16 normals	500 ml of porridge+4.5 g salt	No inhibition vs. 800 mg cimetidine vs. 40 mg omeprazole	GE measured by EIT on three occasions per subject, once with no acid inhibition, once with cimetidine, once with omeprazole	GE $t_{1/2}^1$ was quicker in males than females (control ($P = 0.01$), cimetidine ($P = 0.02$)) In males GE was quickest cimetidine > controls > omeprazole In the females GE was quickest cimetidine > omeprazole > controls In controls, female lag phase > males for semi-solids and liquids ($P = 0.04$, $P = 0.04$) Good agreement for $t_{1/2}^1$ between the two methods ($r = 0.801$, $P < 0.05$)
5	Comparison of GE measured by EIT vs. scintigraphy	8 normals	300 ml consomme+300 ml ^{99m}Tc -tin colloid water	?	GE measured simultaneously by EIT vs. scintigraphy	Good agreement for $t_{1/2}^1$ between the two methods ($r = 0.83$, $P < 0.01$)
5	Comparison of GE measured by EIT vs. dye dilution	10 normals	750 ml 5% aqueous sucrose	?	GE measured simultaneously by EIT vs. dye dilution	Good agreement for $t_{1/2}^1$ between the two methods ($r = 0.83$, $P < 0.01$)
7	Comparison of GE measured by EIT vs. scintigraphy, and effect of acid inhibition	11 normals	1 oxo cube plus water labelled with 2 MBq ^{99m}Tc -DTPA	400 mg cimetidine vs. no acid inhibition	GE measured simultaneously by EIT vs. subject, once with acid inhibition, once without	One subject failed to complete the study to give 20 studies There was good agreement between $t_{1/2}^1$ for EIT and scintigraphy Acid inhibition ($r = 0.87$, $P = 0.001$) Controls ($r = 0.77$, $P = 0.006$)
14	Investigation of the feasibility of using EIT in infants	47 infants	Formula milk (25 ml/kg) or dioralyte	?	GE measured by EIT and gastric residuals	Good agreement at 90 min between GE and gastric residuals in milk fed (16/20) or dioralyte fed (24/27) infants
5	Comparison of GE measured by EIT vs. scintigraphy	8 normals	85 g mashed potato+300 ml ^{99m}Tc -tin colloid water	800 mg cimetidine	GE measured simultaneously by EIT vs. scintigraphy	Good agreement for $t_{1/2}^1$ between the two methods ($r = 0.73$, $P < 0.05$)
7	Comparison of EIT vs. scintigraphy-effect of acid inhibition	8 normals	500 ml of porridge+4.5 g salt labelled with 2 MBq ^{99m}Tc -DTPA	400 mg cimetidine vs. no acid inhibition	GE measured simultaneously by EIT vs. subject, once with acid inhibition, once without	Only 10 studies, showed good agreement between $t_{1/2}^1$ for EIT and scintigraphy Acid inhibition ($P = 0.04$) Controls ($P = 0.04$)

Table 2 Disadvantages of gamma scintigraphy as a measure of gastric emptying.

- Gastric and salivary secretions, which contribute up to 188 ml/h are not included in the measurements.⁴
- Only the radio-labelled phase (usually protein) of the meal is included in the measurement. Most meals are a complex mixture of protein, carbohydrate and emulsion of fat which are likely to be emptied from the stomach at different rates.
- Markers may separate from the protein phase and empty as a non-nutritive liquid resulting in erroneous results; commonly used radio-isotopes lose up to 2–5% of binding from the solid phase per hour.

Materials and methods

Ethics

The study was approved by East London and the City Research Ethics Committee for measurements in normal volunteers and informed consent was obtained.

Study procedure

The test meal was a standard 1 kcal/ml enteral feed (Nutrison Standard Nutricia[®], The Netherlands) radio labelled with technetium tin (Tc99m-Sn) colloid, which predominantly binds to the protein portion of the feed. As the conductivity of stomach tissue (0.5 S/m at 100 Hz)¹⁰ is very similar to that of enteral feed (0.56 S/m) it was necessary to add 1 g (17 mmol) of sodium chloride to each 100 ml of feed to increase conductivity so that the contrast between stomach tissue and feed could be imaged.

The EIT system used was a Sheffield Mark 1 system (Medical Physics Department, Sheffield University). Sixteen leads are placed around the subject's abdomen (Fig. 1), which drive alternating current at 5 mA through an adjacent pair of electrodes while measuring voltage at another adjacent pair. This yields a single transfer impedance measurement. Voltage is measured at all the other pairs of electrodes in turn, then sequential pairs of electrodes act as drive electrodes until all the possible combinations have been measured providing a full set of 208 combinations in 80 ms. This system has a single operating frequency (50 kHz) and a two-dimensional (2D) reconstruction algorithm.⁵ Because variable skin contact impedance at the different electrode sites can introduce error, it was necessary to carry out checks prior to starting EIT recording. The system uses a reciprocity value to check electrode contact. When current is applied across two electrodes a voltage is induced at another two; this voltage should be the same when it is measured in the other (reciprocal) direction. A reciprocity value of greater than 80% is considered acceptable.



Figure 1 Subject positioning for simultaneous EIT and GS measurements.

Gamma camera imaging was carried out using a large field of view camera (General Electric[®], MaxiCamera 400AC, Milwaukee, USA). To allow anterior and posterior abdominal images to be taken the subject lay on an imaging table (Fig. 1) which allowed the camera detector head to be positioned above and below the subject. The camera detector head was rotated manually.

Subjects attended following an overnight fast of at least 8 h. A dose of 300 mg of ranitidine hydrochloride (Ranitic[®]) was taken to suppress gastric acid production. Two hours after ranitidine hydrochloride had been taken a CH8 PVC enteral feeding tube (Nutricia[®], The Netherlands) was passed and gastric placement confirmed by checking the pH of gastric secretions was 5 or less. Blue sensor ECG electrodes (Ambu[®], Denmark) were placed on the subject midway between the sternum and the umbilicus after cleaning the skin with alcohol. Sixteen electrodes were evenly placed starting with electrode 1 at the left of the subject's anterior midline and ending with electrode 16 to the right of the midline. A tape measure was used to ensure that the electrodes were evenly spaced and lay in the same plane. The 17th, or reference electrode, was placed on the right-hand side of the chest above and removed from the

others. Using the EIT system, electrode reciprocity was checked; if a value of less than 80% was obtained, electrodes were re-positioned until the value was greater than 80%. Subjects then lay on the imaging table, positioned under the gamma camera, with their head elevated 30° (Fig. 1).

The administered activity of the radiopharmaceutical (^{99m}Tc -tin colloid) was confirmed by drawing the required volume into a 2 ml syringe and measuring the dose using a radiation dose calibrator. The sodium chloride followed by 10 MBq of Tc-tin colloid was added to the enteral feed immediately prior to infusion. Prior to starting the enteral feed, the EIT reference frame was taken while the stomach was empty, and recording of gastric resistivity using EIT was started at 150 cycles/min. Three minutes after EIT recording had commenced 100 ml of enteral feed were rapidly infused into the stomach over 2 min to enable identification of the region of interest (ROI) which represented the stomach. Enteral feed delivery was then set at 100 ml/h using a Micromax enteral feeding pump (Nutricia®, The Netherlands) and changes in resistivity were recorded using the EIT system.

Gamma camera imaging was commenced as soon as the 100 ml bolus of enteral feed was given. The position of electrodes 3 and 13 were marked using a small amount of Tc-tin colloid absorbed onto squares of filter paper attached to the electrodes. This made subsequent analysis easier as there was a static reference point on all the GS images. Pairs of anterior and posterior images of the abdomen were taken initially every 5 min, reduced to 10 min intervals after 60 min of data collection. Simultaneous EIT and scintigraphic data collection continued for 220 min.

The gastric emptying curves obtained by EIT and GS were plotted for each subject.

GS gastric emptying curve

As a meal moves from the fundus to the antrum it moves posteriorly due to the anatomical position of the stomach in the abdomen. If gamma camera images are taken only from the anterior position this movement increases the thickness of tissue between the stomach and the camera causing attenuation of gamma radiation and a decrease in the number of counts detected externally. Tissue attenuation is minimised by taking pairs of anterior and posterior images and using the geometric mean of the counts for each pair of images to plot the gastric emptying curve.

The ROI that represented the stomach was delineated using the first image, taken immediately after the 100 ml bolus of feed. Subsequent images were synchronised using the markers that had been

placed on electrodes 3 and 13, and the ROI identified excluding any feed that had passed into the small intestine. Once the ROI had been identified on all pairs of anterior and posterior GS images, counts were corrected for decay and the geometric mean of the counts was calculated. GS gastric emptying curves were obtained by plotting the geometric mean over time and were expressed as the corrected number of counts against time.

EIT gastric emptying curve

As with GS, the ROI was identified following a rapid infusion of 100 ml bolus of highly conductive feed which caused a large change in impedance within the stomach. A gastric emptying curve was plotted by back projecting changes in impedance in the ROI over time against the reference frame, which had been taken when the stomach was empty.

Statistical analysis

EIT curves were smoothed using “smooth 4253h” in Stata¹¹ a robust non-linear compound smoother. The smoothed and unsmoothed EIT curves from subject 1 are shown in Fig. 2. EIT gastric emptying curves were expressed as percentage change in gastric impedance against time.

Volume-adjusted curves

For each pair of EIT and GS curves it was assumed that the curves would coincide at the time point when each method was measuring the initial 100 ml bolus and that subsequent differences would be as a result of differences in gastric volume due to gastric secretions. For each pair of curves, the y-axis was adjusted linearly, the factor being established by eye, so that they coincided at the point when the 100 ml bolus was given and the

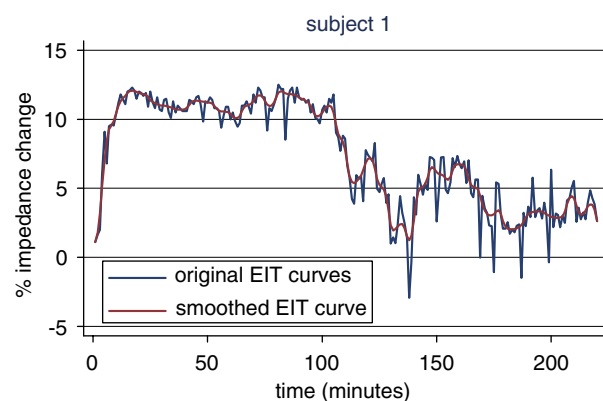


Figure 2 EIT gastric emptying curves showing the original curve and the smoothed curve.

value of the y-axis at this point was adjusted to 100 ml.

Comparison of EIT and GS

Differences in gastric emptying patterns due to acid secretion. As EIT measures total gastric content including gastric acid and saliva and GS does not, volunteers had their gastric acid production suppressed. However, Ranitic® suppresses approximately 60% of acid secretion and does not suppress salivary production. The rate of saliva and gastric acid production has been estimated at 188 ml/h⁴ (63 ml/h saliva, 125 ml/h gastric acid). While these figures are unlikely to accurately reflect the physiological salivary and gastric acid response to nasogastric feeding, a 60% reduction in gastric acid secretion would result in a production rate of 113 ml/h (63 ml saliva, 50 ml gastric acid). As the volume measured by GS was 100 ml/h infusion of enteral feed, and the volume measured by EIT was 100 ml/h infusion of enteral feed plus secretion of gastric acid and saliva, the steady-state gastric volume measured by EIT was predicted to be greater and possibly double that measured by GS.

Potential effect of the 100 ml bolus on gastric emptying patterns. As the $t_{1/2}^1$ for enteral feed is approximately 90 min, and a 100 ml bolus was given prior to starting the continuous infusion, it was possible that this 100 ml bolus would elicit a different gastric emptying pattern to that of the continuous infusion. To establish whether the initial bolus had an effect on gastric emptying, using the original GS and EIT curves, the ratio of area under the curve (AUC) measured by EIT and GS was calculated for the first 90 min and then from 90 min

to the end of the study. Thus, the ratio for the first 90 min ($AUC^{90}_{EIT}:AUC^{90}_{GS}$) and the ratio from 90 to 220 min ($AUC^{220}_{EIT}:AUC^{220}_{GS}$) were compared using a paired *t*-test to identify any difference.

Results

Ten volunteers (five male and five female) were recruited; subject demographics are shown in Table 3. None of the volunteers had any history of gastrointestinal disease and none were constipated. Two subjects had taken medication in the previous 24 h; subject 2 had taken 400 mg of ibuprofen 18 h prior to the study and subject 6 had taken a Mometasone nasal spray for hay fever 15 h prior to the study. Nasogastric tubes were successfully passed into the stomach in all subjects, confirmed by a gastric aspirate pH of less than 5. Gastric emptying curves were obtained in all 10 subjects by EIT but in only eight subjects by GS. In two subjects (subjects 9 and 10), it was not possible to identify the ROI in the GS study due to the presence of radio-labelled feed in the small intestine within 5 min of giving the 100 ml bolus, which obscured the ROI. Therefore, eight paired studies were available for comparison.

Pairs of volume-adjusted curves for the eight subjects are shown (Fig. 3) and comparisons of (a) the pattern of emptying and (b) the estimated gastric volumes trends are summarised in Table 4. For subjects 1, 3 and 8 visual agreement was good as volumes and patterns of emptying and filling appeared similar on both curves. In subject 7, patterns and volumes were similar except for a small decrease in impedance between 100 and

Table 3 Subject demographics for validity studies.

Subject ID	Sex	Age (years)	BMI (m/kg ²)	Medication	Time since last meal (h)	Time since last bowel movement (h)
1	F	27	19	No	10	<12
2	M	57	23	Ibuprofen	8	12–24
3	M	30	19	No	8	<12
4	M	29	25	No	11	12–24
5	F	28	22	No	12	<12
6	M	24	25	Mometasone	11	<12
7	M	37	22	No	12	<12
8	F	27	21	No	12	12–24
9	F	26	22	No	8	<12
10	F	33	24	No	12	12–24
Mean		32	22		10.4	
SD		9.6	2.2			
Range		24–57	19–25		8–12	

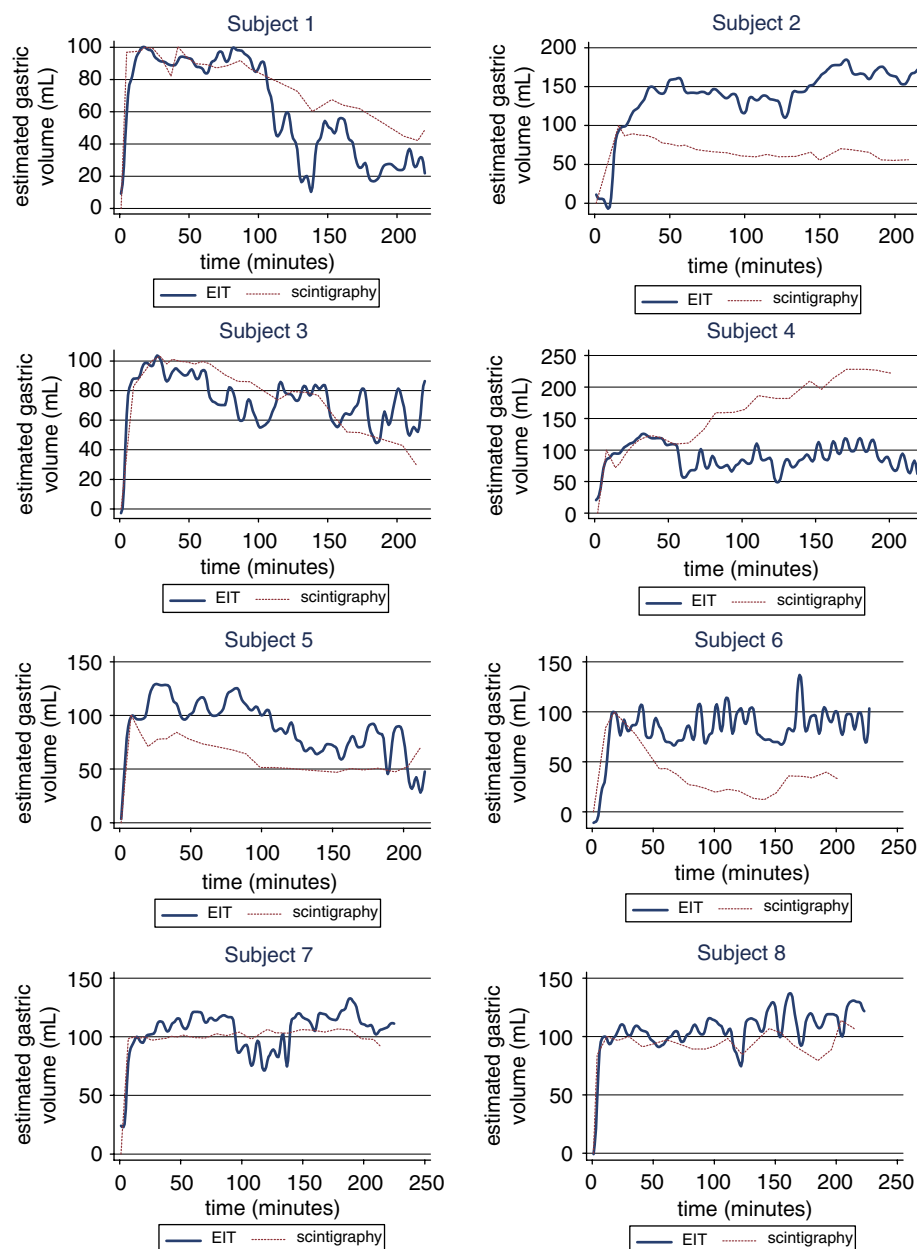


Figure 3 Volume-adjusted EIT and GS curves.

150 min. For subjects 2, 5 and 6, the volume estimated by EIT was larger than that measured by GS; and patterns of emptying and filling were similar in subjects 2 and 5. For subject 6, following the administration of the 100 ml bolus there was an initial emptying pattern on the GS curve that was not seen on the EIT curve, then both curves showed a similar steady state. Subject 4 is anomalous in that the GS curve showed a steady increase in gastric volume throughout the study suggestive of delayed gastric emptying yet the subject had no signs of nausea, vomiting or bloating and the EIT curve showed a steady state suggesting that emptying rate was close to the rate of infusion.

Differences in patterns of gastric emptying due to acid production

Examination of volume-adjusted curves (Fig. 3, Table 4) shows subjects 2, 5 and 6 had larger gastric volumes measured by EIT than GS, subjects 1, 3, 7 and 8 similar volumes and in subject 4 the volume measured by GS was greater and increased with time. Using AUC, for the whole group the volume measured by EIT was 1.34 (SEM = 0.22) times greater than measured by GS (Table 5). However, when subject 4, whose scintigraphy result is clearly anomalous, was excluded the volume measured by EIT was 1.45 (SEM = 0.23) times greater than measured by GS.

Effect of the 100 ml bolus on gastric emptying patterns

Comparison of the difference in the ratios of gastric emptying measured by EIT:GS using the original curves for the first 90 min and from 90 to 220 min found no significant difference; mean difference in ratio = 0.26 ($P = 0.50$, 95% CI -1.11 to 0.60).

General patterns of gastric emptying

Figure 4 shows the gastric emptying curves for the eight volunteers measured by EIT, GS and typical EIT gastric emptying curve showing the pattern of gastric emptying when feed is administered as a bolus. Compared with bolus infusion, where the stomach fills but then empties rapidly, continuous infusion leads to a constant presence of feed in the

stomach. Volume-adjusted curves suggest that the steady-state gastric volume was in the region of 50–125 mL.

Discussion

EIT has previously been extensively validated as a method for measuring gastric emptying of both solid and liquid meals administered as boluses (Table 1). Interpretation of gastric emptying of liquid meals given as a continuous infusion poses more problems whether measured by EIT or GS as there is no obvious measurable end point such as gastric emptying $t_{1/2}^1$, and so patterns of emptying and filling must be interpreted. Although we are aware of one publication¹² which used¹³ carbon labelled acetate to monitor gastric emptying during continuous nasogastric infusion, this is the first direct investigation of gastric emptying patterns during continuous enteral feeding in adults.

When pairs of gastric emptying curves are compared for general patterns of emptying and filling there is good visual agreement in five subjects (1, 2, 3, 5 and 8), although in subjects 2 and 5 the volumes are different. In a further two subjects (6 and 7), curves were similar with the exception of one section of the gastric emptying curve. There is only one subject (subject 4) whose EIT and GS curves appear completely different. This subject shows a clearly anomalous scintigraphy result because scintillation counts increased over time yet there was no evidence of delayed gastric emptying (bloating, vomiting abnormal EIT curve). Rather than exclude this subject as a technical error, we have chosen to include these data for completeness and to demonstrate that no true gold standard exists. Scintigraphy, though well tried and accepted, it self carries inaccuracies.

Table 4 Comparison of gastric emptying curves in each subject.

Subject	Patterns of emptying and filling	Estimated gastric volume
1	Similar	Similar
2	Similar	EIT > GS
3	Similar	Similar
4	Different	GS > EIT
5	Similar	EIT > GS
6	Small discrepancy*	EIT > GS
7	Small discrepancy†	Similar
8	Similar	Similar

*Discrepancy can be explained by gastric acid secretory response.

†Small difference in pattern of gastric emptying between the curves.

Table 5 Difference in volume measured by EIT and GS calculated from area under the curve using volume-adjusted curves.

	AUC total EIT (mL)	AUC total GS (mL)	AUC EIT/AUC GS
Subject 1	13,492	16,363	0.82
Subject 2	29,653	13,770	2.15
Subject 3	16,177	15,850	1.02
Subject 4	19,484	32,081	0.61
Subject 5	19,070	12,783	1.49
Subject 6	18,799	7824	2.40
Subject 7	23,568	21,352	1.10
Subject 8	23,660	20,080	1.17
Mean (SEM)	20,488 (1778)	17,512 (2568)	1.34 (0.22)
Mean (excluding subject 4) (SEM)	20,631 (2046)	15,431 (1728)	1.45 (0.23)

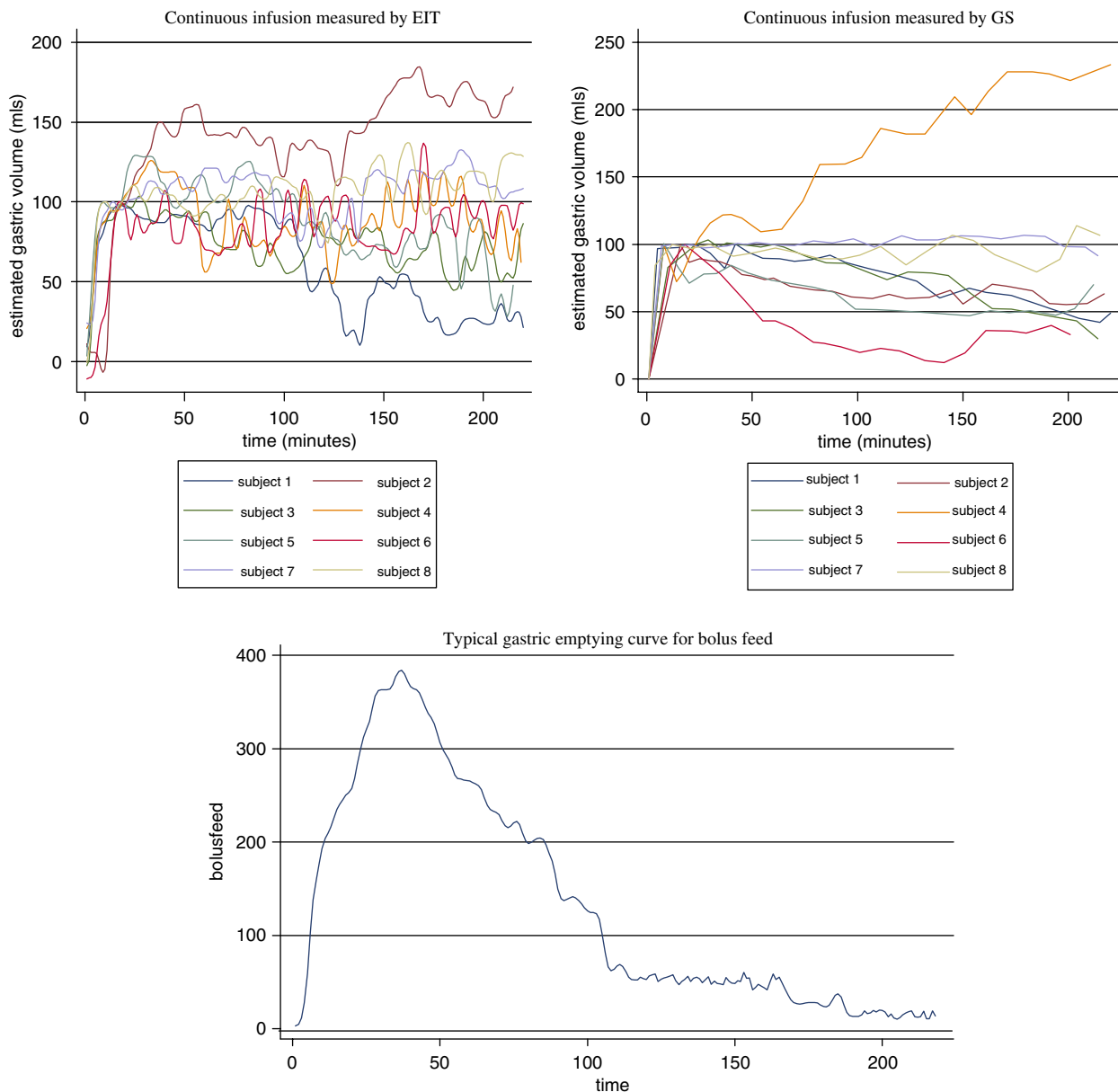


Figure 4 Gastric emptying curves for continuous infusion of enteral feed (upper panel) and bolus feed (lower panel).

Curves were volume-adjusted during an initial 100 ml bolus to allow the relative volumes measured by each method to be compared. It must be stressed that volumes estimated from both EIT and GS curves are approximations; when current is applied to the abdomen it moves in three dimensions but changes in impedance are reconstructed using a 2D algorithm. Both EIT and GS have poor resolution which causes blurring so identification of the exact position of the stomach boundary is difficult so results should be interpreted with caution.

Nonetheless, volume adjustment of the curves allowed us to examine whether the EIT measurements, which included saliva and gastric secretions, were greater than those obtained by GS,

which images only the labelled portion of the feed. The value of AUC measured by EIT divided by GS was expected to be approximately 2 and found to be 1.34 or 1.45 when an outlier was excluded (Table 5). This larger value obtained by EIT was consistent with there being a secretory response which had been variably modified by the ranitidine.

Because the conductivity of enteral feed (0.56 S/m) is very similar to that of stomach muscle (0.52 S/m at 100 Hz)¹⁰ it was necessary to add sodium chloride to the feed to increase its conductivity. As standard enteral feed has a sodium content of 4 mmol/100 ml, the addition of 17 mmol/100 ml of feed represents a significant change. While this additional sodium does not affect gastric emptying

patterns,¹³ it limits the potential repeated use of this EIT system.

Was comparison with GS the most appropriate method to validate EIT? In previous studies, when EIT and GS were compared for bolus administration of liquids (results expressed as gastric emptying $t_{1/2}^1$), the two methods agreed so long as gastric secretions were inhibited.^{5,7} The advantage of using $t_{1/2}^1$ is that if errors (which can under or overestimate gastric volume) are present, they are present at peak volume and half volume and so cancel each other out; but when one method is compared with another, errors will remain and may exaggerate differences. Thus, when pairs of EIT and GS curves are closely compared for patterns of emptying and filling, small differences are seen but general patterns are similar and volume differences can be explained by a secretory response.

As there are no published data available on gastric volume or gastric emptying patterns during continuous nasogastric infusion of enteral feed, the general patterns obtained are of major interest. While our methodology can be criticised for measuring a combination of bolus and continuous infusion, it would not have been possible to measure gastric emptying by either EIT or GS had the initial bolus not been given, as it would not have been possible to identify the ROI or obtain a gastric emptying curve. There was no evidence that the initial bolus affected gastric emptying patterns. Although it would appear that patterns of emptying and filling are complicated and variable between individuals, the prevailing pattern was a steady state at a volume of approximately 50–125 ml.

In conclusion, there was general agreement between patterns of gastric emptying measured by EIT and GS in seven out of eight volunteers, and where differences occurred they could be explained by gastric and salivary secretions. While 2D EIT cannot be expected to provide an accurate estimate of gastric volume during continuous nasogastric infusion, it does provide the potential to see patterns of gastric emptying, particularly volume increases and decreases over time as the net result of gastric secretion vs. emptying. Further, development of EIT to avoid the need for the addition of sodium chloride and to improve sensitivity is needed to make this technique a useful tool for monitoring gastric emptying in patients at risk of gastroparesis. Despite the limitations of both EIT and GS, they have provided data on gastric emptying patterns during continuous nasogastric infusion. Such data may be of use

in the future to develop more evidence based protocols for monitoring enteral feed tolerance in patients at risk of gastroparesis.

Acknowledgements

NHS Research and Development funded the salary for CS. Nutricia Clinical Care funded volunteer expenses. All authors designed the study. CS and MK collected the data. CS analysed the data. EH contributed to the statistical analysis. CS wrote the manuscript with critical review from JP-T and DE. None of the authors declared a conflict of interest.

References

- McClave SA, et al. Use of residual volume as a marker for enteral feeding intolerance: prospective blinded comparison with physical examination and radiographic findings. *J Parenter Enteral Nutr* 1992;16(2):99–105.
- McClave SA, Snider HL. Clinical use of gastric residual volumes as a monitor for patients on enteral tube feeding. *J Parenter Enteral Nutr* 2002;26(6 Suppl):S43–8.
- Burd RS, Lentz CW. The limitations of using gastric residual volumes to monitor enteral feedings: a mathematical model. *Nutr Clin Practice* 2001;16(6):349–56.
- Lin HC, Van Citters GW. Stopping enteral feeding for arbitrary gastric residual volume may not be physiologically sound: results of a computer simulation model. *J Parenter Enteral Nutr* 1997;21(5):286–9.
- Avill R, et al. Applied potential tomography. A new noninvasive technique for measuring gastric emptying. *Gastroenterology* 1987;92(4):1019–26.
- Mangnall YF, et al. Applied potential tomography. Noninvasive method for measuring gastric emptying of a solid test meal. *Dig Dis Sci* 1991;36(12):1680–4.
- Wright JW. *The effect of intraluminal content on gastrointestinal motility in man*. Nottingham: University of Nottingham; 1995 [p. 98].
- Akkermans LM, van Isselt JW. Gastric motility and emptying studies with radionuclides in research and clinical settings. *Dig Dis Sci* 1994;39(12 Suppl):95S–6S.
- Vantrappen G. Methods to study gastric emptying. *Dig Dis Sci* 1994;39(12 Suppl):91S–4S.
- Gabriel C, Gabriel S. *Compilation of the dielectric properties of body tissues at RF and microwave frequencies*. <www.brooks.af.mil/AFRL/HED/hedr/reports/dielectric/home.html>.
- Stata Corporation. *Stata 8*. Texas: Stata PC; 2003.
- Chiolero RL, et al. Labeled acetate to assess intestinal absorption in critically ill patients. *Crit Care Med* 2003; 31(3):853–7.
- Soulsby C, Yazaki E, Evans DF, Powell-Tuck J. Can electrical impedance tomographic spectroscopy (EITS) be used to measure enteral feed tolerance? *Clin Nutr* 2002;21:48.
- Nour S, et al. Applied potential tomography in the measurement of gastric emptying in infants. *J Paediatr Gastroenterol Nutr* 1995;20(1):65–72.